Activity #2 – Operators

General Instructions:

1. Create a Java program based on the questions.
2. Follow the exact required output shown in the question.
3. You must use the specific operator required in each number.
4. Do NOT hardcode the final answers (e.g., System.out.println(86); without computation is not allowed).
5. Show proper use of variables and correct operator usage.
6. Follow proper Java syntax, indentation, and naming conventions.
7. Use System.out.println() for displaying output.
8. Screenshot the code and the output.
9. Compile and run your program to verify correctness before submission.
10. Submission Format: Section_LastName_FirstName_OperatorsActivity#2.pdf

Please answer the following questions:

PART I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

Q1.java › public class Q1 › public static void main(String[])

```java
public class Q1 {

    public static void main(String[] args) {
        // Output 1: true
        int a = 12;
        int b = 10;
        boolean result1 = (a | b) > b;
        System.out.println(result1);

        // Output 2: 10
        int c = 14;
        int d = 15;
        int result2 = ((c & d) >> 1) ^ 13;
        System.out.println(result2);

        // Output 3: false
        int e = 6;
        int f = 9;
        boolean result3 = (e ^ f) << 1 == 45;
        System.out.println(result3);

        // Output 4: 15
        int g = 5;
        int h = 10;
        int result4 = g | h;
        System.out.println(result4);
    }
}
```

JAVA — powershell.exe

```
PS D:\Personal Projects\JAVA> java Q1.java
true
10
false
15
PS D:\Personal Projects\JAVA>
```

2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

```java
public class Q2 {

    public static void main(String[] args) {
        // Output 1: true
        int a = 18;
        int b = 9;
        boolean result1 = (a > b) ? true : false;
        System.out.println(result1);

        // Output 2: false
        int c = 7;
        int d = 14;
        boolean result2 = (c == d) ? true : false;
        System.out.println(result2);

        // Output 3: 25
        int e = 50;
        int f = 50;
        int result3 = (e >= f) ? (e / 2) : (e * 2);
        System.out.println(result3);

        // Output 4: 30
        int g = 15;
        int h = 20;
        int result4 = (g != h) ? (g * 2) : (g + 2);
        System.out.println(result4);
    }
```

JAVA — powershell.exe

```
PS D:\Personal Projects\JAVA> java Q2
true
false
25
30
PS D:\Personal Projects\JAVA>
```

3. Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

```java
public class Q3 {

    public static void main(String[] args) {
        // Output 1: Addition: 120
        int a = 95;
        a += 25;
        System.out.println("Addition: " + a);

        // Output 2: Subtraction: 70
        int b = 90;
        b -= 20;
        System.out.println("Subtraction: " + b);

        // Output 3: Multiplication: 144
        int c = 12;
        c *= 12;
        System.out.println("Multiplication: " + c);

        // Output 4: Division: 16
        int d = 256;
        d /= 16;
        System.out.println("Division: " + d);
    }
}
```

JAVA — powershell.exe

```
PS D:\Personal Projects\JAVA> java Q3
Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16
PS D:\Personal Projects\JAVA>
```

PART II:

Program Requirements:

1. Use at least five (5) different operators from the following categories:
   - Arithmetic operators (+, -, *, /, %)
   - Relational operators (>, <, >=, <=, ==, !=)
   - Logical operators (&&, ||, !)
   - Unary operators (++, --)
   - Assignment operators (+=, -=, *=, /=)
   - Bitwise operators (&, |, ^, <<, >>)

2. The program must produce exactly six (6) outputs only.
3. At least one output must include a tricky increment or decrement case, such as:
   - a++ + ++a
   - --b + b++
   - or any similar expression that modifies the variable within the same statement.
4. Do NOT reuse:
   - Any variable values
   - Any variable names
   - Any logic patterns from Part I.
5. The output must contain:
   - At least two (2) boolean results
   - At least four (4) numeric results

Part2.java › public class Part2 › public static void main(String[])

```java
public class Part2 {

    public static void main(String[] args) {
        // bagsak sa quiz kaya i took extra curriculars.
        int score = 4;
        int bonusPoints = score++ + ++score;
        System.out.println("Bonus Points Earned: " + bonusPoints);

        // minus points pag late nagpasa
        int rawScore = 100;
        rawScore -= 37;
        rawScore %= 20;
        System.out.println("Leftover Score After Penalty: " + rawScore);

        // bagsak sa midterms kaya dasal-dasal nalang
        int writtenScore = 55;
        int passingMark = 30;
        int oralScore = 90;
        int failMark = 45;
        boolean passedBoth =
            (writtenScore > passingMark) && (oralScore != failMark);
        System.out.println("Passed Both Exams: " + passedBoth);

        // nag compare kami ng grade ng katabi ko kasi parehas kami ng sagot
        int student1Grade = 78;
        int student2Grade = 78;
        boolean isDifferent = !(student1Grade == student2Grade);
        System.out.println("Grades Are Different: " + isDifferent);

        // add 2 grades
        int grade1 = 51;
        int grade2 = 15;
        int encodedGrade = (grade1 ^ grade2) << 1;
        System.out.println("Encoded Final Grade: " + encodedGrade);

        // make up quizzes
        int quizScore = 9;
        int retakeTotal = --quizScore + quizScore--;
        retakeTotal *= 2;
        System.out.println("Weighted Retake Score: " + retakeTotal);
    }
}
```

OUTPUT:

```
JAVA — powershell.exe
PS D:\Personal Projects\JAVA> javac Part2.java
PS D:\Personal Projects\JAVA> java Part2
Bonus Points Earned: 10
Leftover Score After Penalty: 3
Passed Both Exams: true
Grades Are Different: false
Encoded Final Grade: 120
Weighted Retake Score: 32
PS D:\Personal Projects\JAVA>
```